

Understanding Switch MAC Address Learning and Forwarding

Objective:

To observe how a Cisco switch dynamically builds its MAC address table and to understand the difference between broadcast traffic (ARP) and known unicast traffic.

CASE 1: SINGLE SWITCH CONFIGURATION

Part 1: Building the physical topology

1. **Add the Switch:** Open a blank Cisco Packet Tracer workspace. From the bottom-left menu, select **Switches** and drag a **2960** switch into the center of your workspace.
2. **Add the End Devices:** Select **End Devices** and drag **three PCs** onto the workspace around the switch. Name them **PC-A**, **PC-B**, and **PC-C**.
3. **Cable the Network:** Select the **Connections** menu (the lightning bolt) and choose the **Copper Straight-Through** cable (solid black line).
 - Connect **PC-A** (FastEthernet0) to the Switch on port **FastEthernet0/1**.
 - Connect **PC-B** (FastEthernet0) to the Switch on port **FastEthernet0/2**.
 - Connect **PC-C** (FastEthernet0) to the Switch on port **FastEthernet0/3**.
4. **Wait for Convergence:** You will notice the link lights on the switch ports start as amber (orange) and eventually turn green. Wait until all link lights are green before proceeding.

Part 2: Configuring IP Addresses

Switches operate at Layer 2 (MAC addresses), but to generate traffic, our PCs need Layer 3 (IP) addresses to talk to each other.

1. Click **PC-A** > **Desktop** tab > **IP Configuration**. Set the IPv4 Address to **192.168.1.10** and the Subnet Mask to **255.255.255.0**.
2. Click **PC-B** > **Desktop** tab > **IP Configuration**. Set the IPv4 Address to **192.168.1.20** and the Subnet Mask to **255.255.255.0**.
3. Click **PC-C** > **Desktop** tab > **IP Configuration**. Set the IPv4 Address to **192.168.1.30** and the Subnet Mask to **255.255.255.0**.

Part 3: Inspecting the Empty Switch

Before we send any traffic, let's look at the switch's memory.

1. Click on the Switch and go to the CLI tab.
2. Press Enter to get started, then type the following commands to enter Privileged EXEC mode and view the MAC address table:

```
enable
show mac address-table
```

3. **Observation:** The table should be completely empty (or only show a few internal CPU/VLAN entries depending on the Packet Tracer version). Because the PCs haven't sent any data yet, the switch does not know their MAC addresses or which ports they are plugged into.

Part 4: Watching the Switch learn (Simulation Mode)

Now, we will freeze time and watch the switch handle a brand-new connection request.

1. **Enter Simulation Mode:** In the bottom-right corner of Packet Tracer, click the **Simulation** tab (the stopwatch icon).
2. **Filter the Traffic:** Click **Show All/None** at the bottom of the Simulation Panel to clear the checkboxes. Then click **Edit Filters**, go to the **IPv4** tab, and check **ICMP** and **ARP**.
3. **Prepare the Ping:** Open PC-A, go to the **Command Prompt**, and type: `ping 192.168.1.20` (Do not press Enter yet!)
4. **Execute and Step Forward:** Press **Enter** on the ping command. You will see two envelopes appear next to PC-A.
 - **The ARP Packet:** Since PC-A knows PC-B's IP address but not its physical MAC address, it creates an ARP (Address Resolution Protocol) broadcast to ask, "Who has 192.168.1.20?"
5. **Click the 'Capture / Forward' button** (right arrow with a vertical line in the Simulation Panel):
 - The ARP packet travels from PC-A to the Switch.
 - **Crucial Learning Moment:** The moment the switch receives this frame on port Fa0/1, it looks at the Source MAC Address on the frame. It now permanently learns that PC-A's MAC address is located on port Fa0/1.
6. **Click 'Capture / Forward' again:**
 - Because the ARP frame has a Destination MAC Address of `FFFF.FFFF.FFFF` (a broadcast), the switch does not know where PC-B is. Therefore, it **floods** the packet out of every active port except the one it came from. You will see the packet go to both PC-B and PC-C.
7. **Click 'Capture / Forward' again:**
 - PC-C drops the packet (red X) because its IP is not .20.
 - PC-B accepts it and generates an ARP Reply.
8. **Click 'Capture / Forward' again:**
 - The ARP Reply travels from PC-B back to the Switch.
 - **Crucial Learning Moment:** The switch looks at the source of this reply. It now learns that PC-B's MAC address is located on port Fa0/2.
9. **Click 'Capture / Forward' again:**
 - Now, the switch needs to forward this reply to PC-A. Because it *already learned* that PC-A is on port Fa0/1 (back in step 5), it sends the frame **directly** to PC-A. It does not flood it to PC-C. This is known as **Unicast** forwarding.
10. **The ICMP Ping:** Keep clicking forward. Now that MAC addresses are known, you will see the actual ICMP Ping packets travel directly between PC-A and PC-B without ever bothering PC-C.

Part 5: Verifying the learning process

Let's go back into the switch and verify what it learned during that simulation.

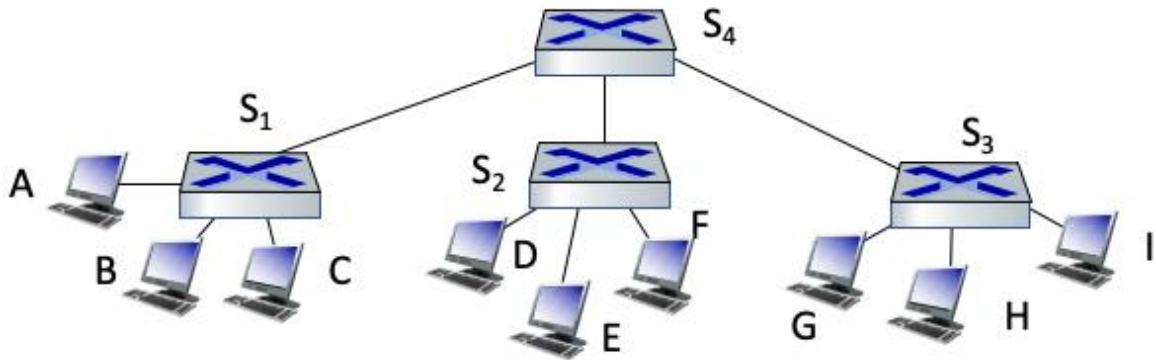
1. Switch back to **Realtime** mode (bottom right corner).
2. Click on the **Switch**, go back to the **CLI**, and run the command again:

```
show mac address-table
```

3. **Observation:** You should now see two distinct entries.

- One MAC address mapped to Fa0/1 (PC-A).
- One MAC address mapped to Fa0/2 (PC-B).
- Note that Fa0/3 is still missing because PC-C never sent any traffic into the switch.

CASE 2: MULTIPLE SWITCH CONFIGURATION



Part 1: Building the physical topology:

1. Place the Switches: Drag four 2960 Switches onto your Packet Tracer workspace.
 - Name the top, central switch S4.
 - Name the bottom three switches S1 (left), S2 (middle), and S3 (right).
2. Place the End Devices: Drag nine PCs onto the workspace and arrange them in groups of three under the access switches. Name them PC-A through PC-I.
3. Cable the Inter-Switch Links (Cross-Over): Use Copper Cross-Over cables to connect the switches together.
 - Connect S4 (FastEthernet0/1) to S1 (FastEthernet0/24).
 - Connect S4 (FastEthernet0/2) to S2 (FastEthernet0/24).
 - Connect S4 (FastEthernet0/3) to S3 (FastEthernet0/24).
4. Cable the PCs (Straight-Through): Use Copper Straight-Through cables to connect the computers to their local switches.
 - Connect PCs A, B, and C to S1 (FastEthernet0/1, 0/2, 0/3).
 - Connect PCs D, E, and F to S2 (FastEthernet0/1, 0/2, 0/3).
 - Connect PCs G, H, and I to S3 (FastEthernet0/1, 0/2, 0/3).
5. Wait for Convergence: Wait a few moments for the Spanning Tree Protocol to finish calculating. All link lights should turn from amber to green.

Part 2: Configure IP addresses:

To allow the PCs to generate traffic and trigger the MAC learning process, we will put them all in the same local subnet (192.168.1.0/24).

1. Open each PC, navigate to the **Desktop** tab, and select **IP Configuration**.
2. Assign the following IP addresses (Subnet Mask for all is 255.255.255.0):
 - **S1 PCs:** PC-A (192.168.1.1), PC-B (192.168.1.2), PC-C (192.168.1.3)

- **S2 PCs:** PC-D (192.168.1.4), PC-E (192.168.1.5), PC-F (192.168.1.6)
- **S3 PCs:** PC-G (192.168.1.7), PC-H (192.168.1.8), PC-I (192.168.1.9)

Part 3: Inspecting core Switch 4:

Let's look at the central switch before we send any data across the network.

1. Click on **S4** and go to the **CLI** tab.
2. Enter Privileged EXEC mode and view the MAC table:

```
enable
show mac address-table
```

3. **Observation:** The table will likely be empty, or you might see a few MAC addresses pointing to the switch interconnects (due to background protocols like STP or CDP). However, S4 has *no idea* where PC-A or PC-I are located yet.

Part 4: Generating cross-network traffic:

Now, we will send a ping from the far left of the network (PC-A) to the far right (PC-I) and observe how the switches learn the path.

1. Open **PC-A**, go to the **Command Prompt**, and ping PC-I: `ping 192.168.1.9`
2. **What happens under the hood:**
 - **Step 1 (ARP Broadcast):** PC-A sends a broadcast asking for the MAC of 192.168.1.9.
 - **Step 2 (S1 Learns):** Switch **S1** receives this frame, learns PC-A's MAC is on port Fa0/1, and floods the broadcast out to PCs B, C, and up to S4.
 - **Step 3 (S4 Learns):** Switch **S4** receives the broadcast on port Fa0/1. It now learns that PC-A can be reached by going out port Fa0/1. It floods the broadcast down to S2 and S3.
 - **Step 4 (S3 Learns):** Switch **S3** receives it, learns PC-A is up its link to S4, and floods it to G, H, and I.
 - **Step 5 (The Reply):** PC-I replies directly to PC-A. As this unicast reply travels back, **S3**, **S4**, and **S1** all examine the source MAC of the reply and permanently learn where PC-I is located.

Part 4: Verifying Switch 4 MAC Address table:

Let's go back to the central switch and see what it learned from that single ping.

1. Click on **S4**, go to the **CLI**, and run the command again:

```
show mac address-table
```

Observation: You should now see two important entries in S4's table:

- You will see the MAC address of **PC-A** mapped to **FastEthernet0/1** (the link pointing to S1).
- You will see the MAC address of **PC-I** mapped to **FastEthernet0/3** (the link pointing to S3).

Central switches (like S4) do not need to be directly plugged into a computer to learn its MAC address. They learn the physical addresses of remote devices simply by listening to the source MAC addresses of the traffic flowing through their interconnecting trunk links.